DSA06 - Data Handling and Visualization
Activity 5: Normality Detection using Q-Q Plots | Slot D

R.No: 12 | Reg. No: 192424277 | Name: GUNA P

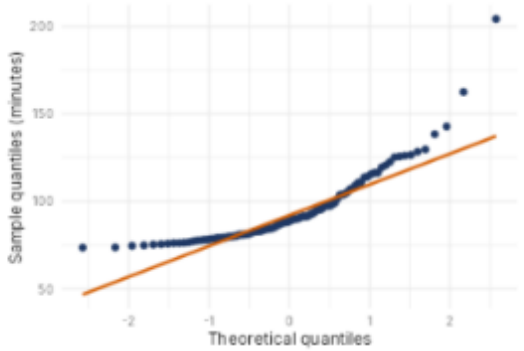
Question Paper

DSA06 - Data Handling and Visualization | Activity 5: Normality Detection using Q-Q Plots | Slot D

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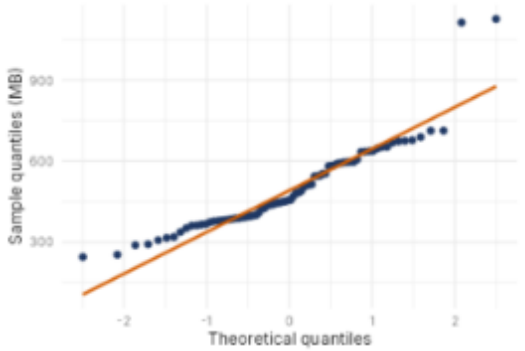
Each chart is a normal Q-Q plot made in R (stat_qq and stat_qq_line; red line = perfect normal fit). Part A (16 marks): for every chart decide whether the data are normal and name the shape from this list: Normal, Right-skewed, Left-skewed, Heavy-tailed, Light-tailed, Bimodal, Normal with outlier(s); then cite the evidence you see.

Chart 1: daily study time (minutes), n = 100



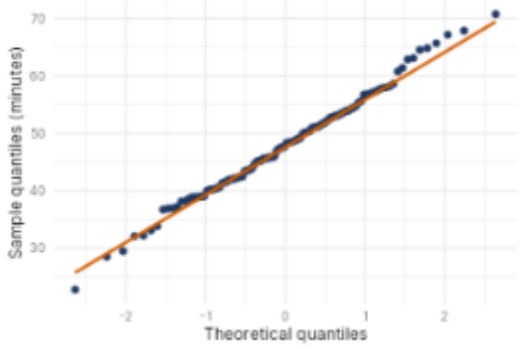
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 2: mobile data used per day (MB), n = 80



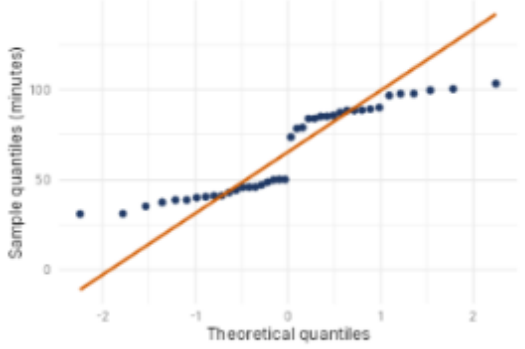
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 3: commute time (minutes), n = 120



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 4: daily study time (minutes), n = 40



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Part B (4 marks)

B1. Chart 1 departs from normality. Name the departure and state the remedy you would try before using mean-and-SD methods (for example log transform, square, split the data, robust methods). [2]

B2. Fill the blanks to draw a Q-Q plot with its reference line for a data frame df with a column value: ggplot(df, aes(sample = value)) + _____ + _____ [2]

Question (typed)

Each chart is a normal Q-Q plot made in R (stat_qq and stat_qq_line; red line = perfect normal fit). Part A (16 marks): for every chart decide whether the data are normal and name the shape from this list: Normal, Right-skewed, Left-skewed, Heavy-tailed, Light-tailed, Bimodal, Normal with outlier(s); then cite the evidence you see.

How to read a normal Q-Q plot

- X-axis = theoretical quantiles (what a perfect normal sample would give). Y-axis = sample quantiles (the real data).
- Points lying on the red line means the data are normal.
- Curved (convex) bend with the upper tail above the line means right-skewed; the mirror image means left-skewed.
- Both ends bending away in opposite directions (S-shape) means heavy-tailed (or light-tailed if the S is reversed).
- Points on the line except one or two far away means normal with outlier(s).
- Flat steps with a gap or jump in the middle means two clusters, i.e. bimodal.

Part A Answers (16 marks)

Chart 1: daily study time (minutes), n = 100	
Normal? (Yes / No)	No
Shape (from the list)	Right-skewed
Evidence in the plot	**1.** The points do not follow the red line. They form a convex curve that bends upward, like the right side of a bowl. **2.** On the left and in the middle, the points are flat and bunched together at about 70 to 100 minutes. Many students study a similar, fairly short time, so the lower values are compressed. **3.** On the right, the points climb steeply above the red line (about 130 to 160 minutes) and one very large value reaches about 200 minutes, far above where a normal sample of this size would reach.
Why this shape?	Upper-tail points lie above the line while the lower values are squeezed together. That means a long tail on the right side, so the mean is pulled above the median. This is the typical Q-Q pattern for right-skewed data.

Chart 2: mobile data used per day (MB), n = 80	
Normal? (Yes / No)	Yes, the main body of the data is normal, apart from a few outliers
Shape (from the list)	Normal with outlier(s)
Evidence in the plot	**1.** Through the lower and middle part (about 250 to 600 MB) the points sit right on or very close to the red line. **2.** Near the top-right end, a couple of points (about 880 to 900 MB) jump far above the line and are clearly separated from the rest. **3.** The rest of the upper and lower tails still follow the line, so there is no overall curve or S-shape.
Why this shape?	Only isolated points leave the line while the rest of the data agree with it. A skewed or heavy-tailed sample would bend gradually across a whole tail, but here just one or two extreme values stand out, so they are outliers (for example unusually heavy data-usage days).

Chart 3: commute time (minutes), n = 120	
Normal? (Yes / No)	Yes
Shape (from the list)	Normal
Evidence in the plot	**1.** The points lie very close to the red line from the lowest value (about 20 minutes) to the highest (about 70 minutes). **2.** The only deviations are tiny random wiggles on either side of the line, with no consistent bend upward or downward. **3.** There are no gaps, no steps and no isolated extreme points at either end.
Why this shape?	Sample quantiles match theoretical normal quantiles all along the line, with no systematic curve, S-shape, gap or outlier. The data are consistent with a normal distribution, so mean-and-SD methods are fine.

Chart 4: daily study time (minutes), n = 40	
Normal? (Yes / No)	No
Shape (from the list)	Bimodal
Evidence in the plot	**1.** The points make a stair-step or S-shaped pattern that crosses the red line instead of following it. **2.** The lower half forms a flat plateau at about 40 to 50 minutes, so many values are repeated around the same level. **3.** In the middle there is a sudden jump (a gap) from about 50 up to about 80 minutes, and then the upper half forms a second plateau at about 90 to 100 minutes.
Why this shape?	A plateau, a gap and a second plateau mean the data fall into two separate clusters, with a low-study group and a high-study group. Two peaks means bimodal, a mixture of two groups, and not a single normal bell curve.

Part B Answers (4 marks)

B1. Chart 1 departs from normality. Name the departure and state the remedy you would try before using mean-and-SD methods (for example log transform, square, split the data, robust methods). [2]

Departure

Chart 1 departs from normality through **right skewness** (a long right tail). The points bend upward above the red line at the top end, the lower values are bunched, and there is an extreme value of about 200 minutes. Because of this, the mean is inflated and the SD is large, so mean-and-SD methods would give misleading results.

Remedy

I would first try a **log transformation** of the study time.

- Taking log() shrinks large values much more than small ones, which pulls in the long right tail and makes the distribution more symmetric.
- Then redraw the Q-Q plot of the transformed data. If the points now follow the red line, mean-and-SD methods can be used on the log scale.
- If a log transform is not enough, use a square-root transform, or use robust methods (median and IQR, rank-based tests) instead of mean and SD.
- Also check the extreme value (about 200 minutes) for a data-entry error before deciding what to do with it.

```
df$log_value <- log(df$value)

ggplot(df, aes(sample = log_value)) +
  stat_qq() +
  stat_qq_line()
```

B2. Fill the blanks to draw a Q-Q plot with its reference line for a data frame df with a column value: ggplot(df, aes(sample = value)) + _____ + _____ [2]

Answer

First blank: **`stat_qq()`** Second blank: **`stat_qq_line()`**

```
ggplot(df, aes(sample = value)) +
  stat_qq() +
  stat_qq_line()
```

Explanation

- `aes(sample = value)` tells ggplot which column to use for the sample quantiles on the y-axis.

- `stat_qq()` draws the points: each sorted data value plotted against the matching theoretical normal quantile.
- `stat_qq_line()` adds the reference line (the red line in the charts) that shows where the points would fall if the data were perfectly normal.